27/2/2026

Feature Scaling (technique to Standardize the independent feature present in a fixed range)

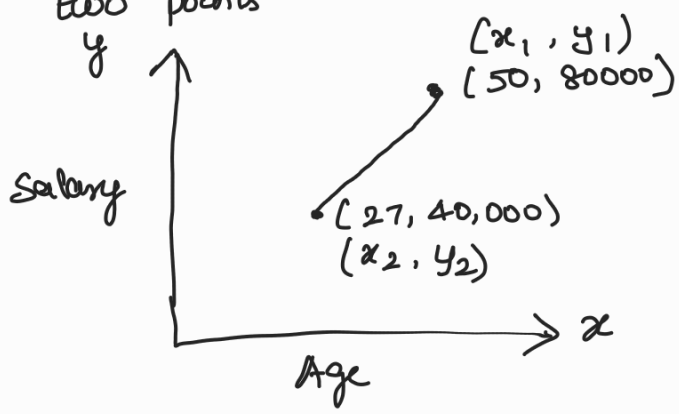
Standardization

Why do we need feature scaling?

Consider a dataset :

Age	Salary
50	80,000
27	40,000
⋮	⋮

Say you are using KNN algorithm: so we predict the o/p based on Distance b/w the two points



$$\text{Eucilidean distance} = \sqrt{(x_2 - x_1)^2 - (y_2 - y_1)^2}$$

$$(x_2 - x_1)^2 = (27 - 50)^2 = 529$$

$$(y_2 - y_1)^2 = (40,000 - 80,000)^2 = 1,60,00,000,000$$

Both values are in 2 different range hence KNN will dominate on one end and the output is not reliable. Hence we use Standardization technique to bring both the values into same range for prediction.

Types of Scaling

- Scaling
  - Standardization
  - Normalization
    - min-max scaling
    - Robust scaling
    - ⋮

Standardization is also called **Z-Score Normalization**

Intuition

Age (x_i)	Salary (y_i)	Age (x_i')	Salary (y_i')
20	10,000		
25	18,000		
27	25,000		
38	40,000		
46	⋮		
⋮			
500 values	500 values		

Standardization converts Age (x_i) into Age (x_i')

$$x_i' = \frac{x_i - \bar{x}}{\sigma}$$

$\bar{x}$ = mean

σ = standard Deviation

say $\bar{x}$ of Age column is 32 and

σ of Age column is 10

$$x_1' = \frac{x_1 - \bar{x}}{\sigma} = \frac{20-32}{10} \qquad \text{|||ly } y_1' = \frac{y_1 - \bar{y}}{\sigma}$$

$$x_2' = \frac{x_2 - \bar{x}}{\sigma} = \frac{25-32}{10} \qquad \vdots$$

$$\vdots \qquad \vdots \qquad \vdots$$

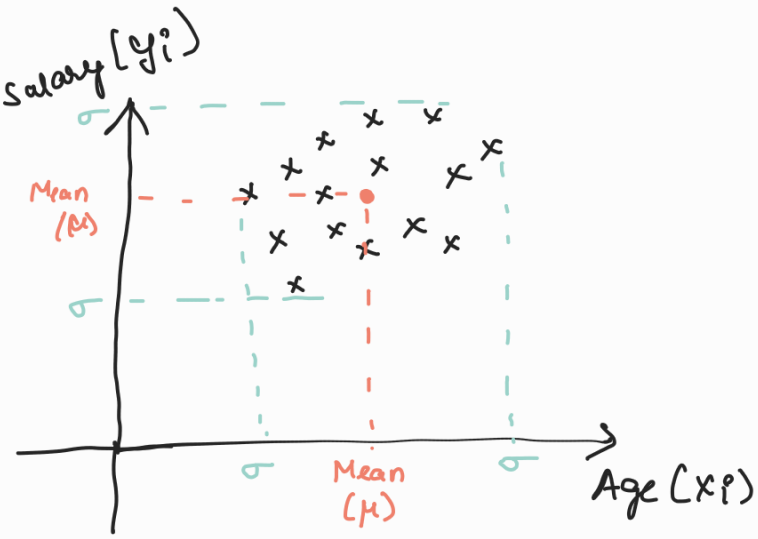
$$x_{500}' = \frac{x_{500} - \bar{x}}{\sigma} = \frac{50-32}{10} \qquad y_{500}' = \frac{y_{500} - \bar{y}}{\sigma}$$

Now we get Age (x_i') and salary (y_i') columns.

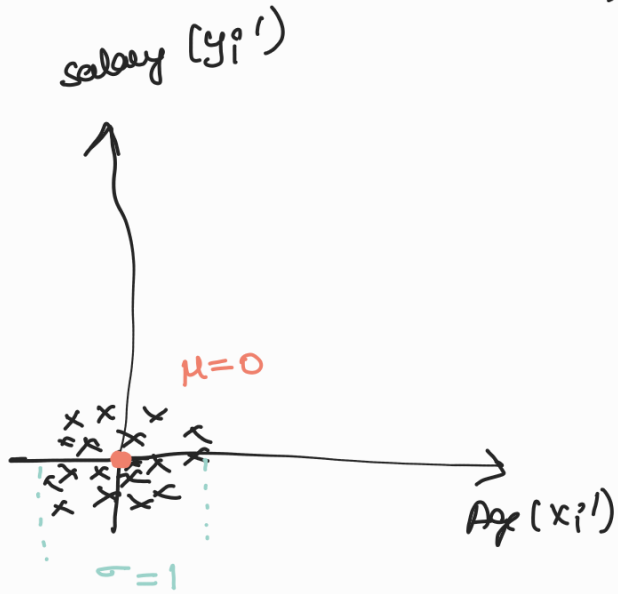
Now the mean and standard deviation of x_i' and y_i' will be 0 and 1 respectively. $\boxed{\mu = 0 \ , \ \sigma = 1}$

Geometric intuition

x_i and y_i (Before standardization)



x_i' and y_i' (After standardization)



Before standardization we can see the mean & SD are in different values. After standardization the mean is centered to zero. This is called mean centering

And Standard deviation $\sigma = 1$. Whatever may be the standard deviation after standardization the spread value i.e., σ will be 1

When to use standardization ?

Algorithm
- K - Means } distance
- KNN } distance
- PCA → maximum variance.
- ANN → Gradient descent is applied
- Gradient descent. (Weight based)

The above algorithm requires standardization.

Decision tree / Random Forest / XGboost / Gradient boost

These algorithm doesn't require stdardization.